

Elamaran M

Full Stack Developer

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Profile Summary

Deeply curious and execution-focused MERN Stack Developer, consistently driven to explore new technologies and bring innovative ideas to improve existing systems. Disciplined and determined to gain practical experience while turning complex ideas into working software.

Projects

MBBS AI Study Assistant (RAG-based Exam Prep Platform)

[Link](#)

- Built a full-stack **RAG-based** study platform that lets students search previous-year MBBS exam questions across multiple universities and retrieve verified textbook references (Robbins Pathology, Harrison's Internal Medicine, Bailey & Love's Surgery).
- Designed a **RAG pipeline** using **Qdrant vector database** for semantic similar-questions search, generating structured, syllabus-aligned AI answers (Definition, Clinical Features, Investigations, Management) grounded in real source excerpts.
- Developed a **Study from PDF** module allowing users to upload their own study material, which is chunked and embedded into the vector store to enable grounded, document-specific Q&A.
- Architected a weighted **MongoDB text search** with automatic regex fallback, ensuring accurate and reliable results even for partial or oddly-phrased queries.

DevMind (AI Agent for Developer Debugging)

[Link](#)

- Built a multi-tool **AI agent** using the **ReAct pattern** where LLM autonomously decides to read GitHub code or search the web, returning exact bug fixes with file name and line number.
- Integrated **Groq tool calling** with two custom tools — `read_github_file` (GitHub REST API) and `search_web` (Tavily API) — where LLM dynamically selects which tool to invoke instead of following hardcoded logic.
- Leveraged **openai/gpt-oss-120b** via Groq's low-latency inference API to ensure reliable multi-step tool calling across the agent loop.
- Secured per-project GitHub token storage with **JWT HTTP-only cookies** and protected routes, ensuring each user's repo and notes are fully isolated in MongoDB.
- Built **three-panel workspace UI** with file tree, agent chat, and auto-saved solutions panel using React and Redux Toolkit for global state management.

Nova-Share (AI-Powered File Management System)

[Link](#)

- Developed a full-stack secure file sharing platform with **JWT authentication**, protected routes, and HTTP-only cookies, ensuring end-to-end security.
- Integrated **Groq's Llama 3.3 70B LLM** to enable AI-powered smart file search and file summarization, demonstrating practical LLM integration.
- Utilized **Cloudinary API** for cloud-based file storage, enabling unique short link sharing, password protection, and configurable file expiry.
- Built a real-time dashboard tracking uploads, downloads, and file statistics using **Redux Toolkit** for state management and MongoDB for persistent storage.
- Deployed frontend on **Vercel** and backend on **Render** with environment-based configuration, CORS handling.

Skills

Frontend: React, JavaScript (ES6+), Tailwind CSS, HTML, CSS
Backend & Database: Node.js, Express.js, MongoDB, Mongoose, JWT, Qdrant (VectorDB)
Tools & Deployment: Git, GitHub, Vercel, Render
AI & LLM Integration: OpenRouter, Groq API

Education

Government College of Engineering – Salem
BE in Mechanical Engineering

2022–2026
CGPA – 7.63